Experiment – 6

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Branch: BE.CSE **Section/Group:** 606/B
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Subject Name: FS Development-II **Subject Code:** 23CSH-309

1. **Aim:** To validate React component behavior by simulating real user interactions and analyzing rendering performance and API behavior.

2. **Objectives:**

- To simulate user behavior using React Testing Library.
- To analyze component re-render cycles using React DevTools Profiler.
- To inspect API responses and network latency using Browser DevTools.
- To evaluate component performance and identify optimization opportunities.
- To monitor network requests and verify correct API communication

3. **Theory:**

React component validation is used to ensure that components behave correctly when users interact with them. Tools like React Testing Library allow developers to simulate real user actions such as clicking buttons or entering input and verify the expected behavior. React DevTools Profiler helps analyze how components render and measure the time taken for each render cycle, which helps in identifying unnecessary re-renders and improving performance. Browser DevTools, especially the Network tab, allows developers to inspect API requests, responses, and loading times during application execution. Together, these tools help developers test functionality, optimize performance, and ensure smooth communication between the frontend and APIs.

4. **Implementation/Code:**

setupTests.js

```
import '@testing-library/jest-dom';
```

App.test.js

```
import { render } from "@testing-library/react";
import App from "./App";

jest.mock("react-router-dom", () => ({
BrowserRouter: ({ children }) => <div>{children}</div>,
Routes: ({ children }) => <div>{children}</div>,
Route: ({ element }) => element,
Link: ({ children }) => <a>{children}</a>,
}));

test("App renders without crashing", () => {
render(<App />);
});
```

App.js

```
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";

import Navbar from "./components/Navbar";
import Footer from "./components/Footer";

import Home from "./pages/Home";
import Contact from "./pages/Contact";
import About from "./pages/About";
import Login from "./pages/Login";
import Register from "./pages/Register";

import "./App.css";

function App() {
return (
<Router>

<Navbar />

<main className="main-content">

<Routes>

<Route path="/" element={<Home />} />

<Route path="/about" element={<About />} />

<Route path="/contact" element={<Contact />} />

<Route path="/login" element={<Login />} />
```

```
<Route path="/register" element={<Register />} />

</Routes>

</main>

<Footer />

</Router>
);
}

export default App;
```

Components.test.js

```
import { render, screen } from "@testing-library/react";
import Navbar from "../components/Navbar";
import Footer from "../components/Footer";
import Hero from "../components/Hero";
import Feature from "../components/Feature";
import Card from "../components/Card";
import InputField from "../components/InputField";

import AboutHeader from "../components/AboutHeader";
import AboutMission from "../components/AboutMission";
import AboutTeam from "../components/AboutTeam";
import AboutStats from "../components/AboutStats";

import Login from "../pages/Login";
import Register from "../pages/Register";
import Home from "../pages/Home";
import Contact from "../pages/Contact";
import About from "../pages/About";

jest.mock("react-router-dom", () => ({
BrowserRouter: ({ children }) => <div>{children}</div>,
Routes: ({ children }) => <div>{children}</div>,
Route: ({ element }) => element,
Link: ({ children }) => <a>{children}</a>,
}));

test("Navbar renders", () => {
render(<Navbar />);
expect(screen.getByText(/home/i)).toBeInTheDocument();
});

test("Footer renders", () => {
render(<Footer />);
```

```
expect(screen.getByText(/2026/i)).toBeInTheDocument();
});

test("Hero renders", () => {
render(<Hero />);
expect(screen.getByText(/welcome/i)).toBeInTheDocument();
});

test("Feature renders", () => {
render(<Feature />);
expect(screen.getByText(/features/i)).toBeInTheDocument();
});

test("Card renders", () => {
render(<Card />);
expect(screen.getByText(/card/i)).toBeInTheDocument();
});

test("InputField renders", () => {
render(<InputField label="Name" type="text" />);
expect(screen.getByText(/name/i)).toBeInTheDocument();
});

test("AboutHeader renders", () => {
render(<AboutHeader />);
});

test("AboutMission renders", () => {
render(<AboutMission />);
});

test("AboutTeam renders", () => {
render(<AboutTeam />);
});

test("AboutStats renders", () => {
render(<AboutStats />);
});

test("Login page renders", () => {
render(<Login />);
expect(screen.getByText(/login/i)).toBeInTheDocument();
});

test("Register page renders", () => {
render(<Register />);
expect(screen.getByText(/register/i)).toBeInTheDocument();
});
```

```
test("Home page renders", () => {
render(<Home />);
});

test("Contact page renders", () => {
render(<Contact />);
});

test("About page renders", () => {
render(<About />);
});
```

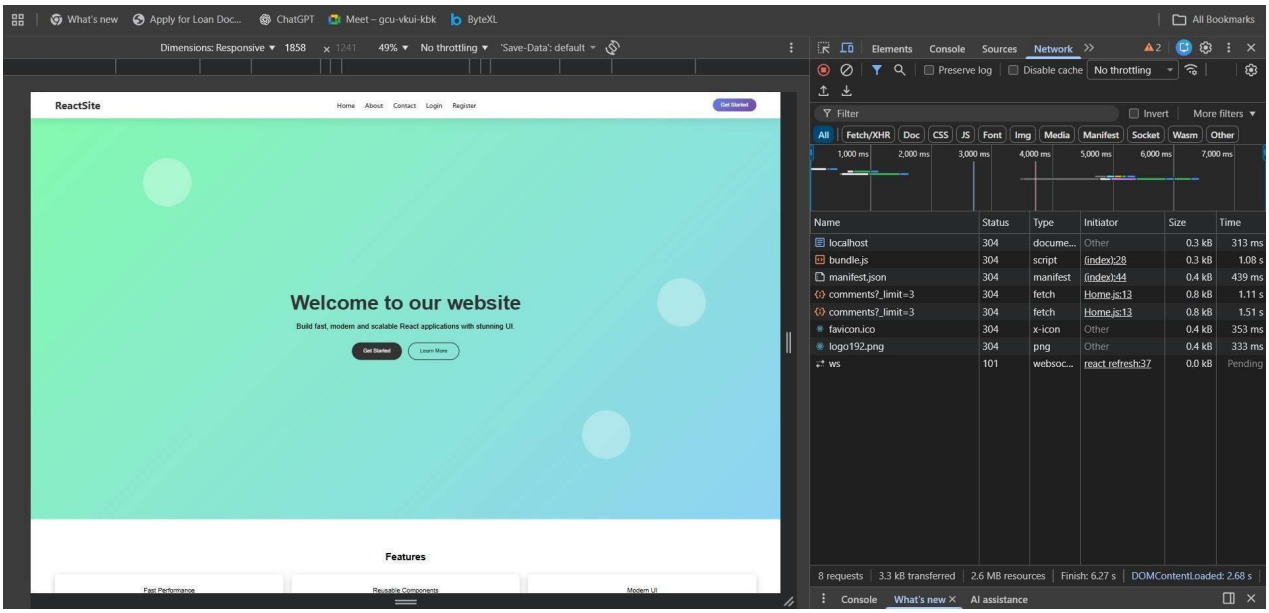
Home.js

```
import Hero from "../components/Hero";
import Feature from "../components/Feature";
import Card from "../components/Card";
import { motion } from "framer-motion";
import { useState, useEffect } from "react";
import "./Home.css";

function Home() {
const [testimonials, setTestimonials] = useState([]);

useEffect(() => {
fetch("https://jsonplaceholder.typicode.com/comments?_limit=3")
.then((res) => res.json())
.then((data) => setTestimonials(data));
}, []);

return (
<div className="home">
<Hero />

<Feature />

{/* Services */}
<section className="services">
<h2>Our Services</h2>

<div className="services-grid">
<Card />
<Card />
<Card />
</div>
</section>

{/* Testimonials from API */}
<section className="testimonials">
```

```
<h2>Client Testimonials</h2>

<div className="testimonial-grid">
{testimonials.map((item) => (
<motion.div
className="testimonial"
key={item.id}
initial={{ opacity: 0, y: 30 }}
animate={{ opacity: 1, y: 0 }}
transition={{ duration: 0.6 }}
>
<h4>{item.email}</h4>
<p>{item.body}</p>
</motion.div>
))}
</div>
</section>
</div>
);
}

export default Home;
```

5. Output :

```
> react-router-project@0.1.0 test
> react-scripts test
 PASS  src/App.test.js (5.994 s)
 PASS  src/tests/components.test.js (6.218 s)

Test Suites: 2 passed, 2 total
Tests:       16 passed, 16 total
Snapshots:   0 total
Time:        13.383 s, estimated 26 s
Ran all test suites related to changed files.
```



6. Learning Outcomes:

- I have learnt to understand how to test React components by simulating user interactions using React Testing Library.
- I have learnt to analyze component rendering performance using the React DevTools Profiler.
- I have learnt to Understand how to inspect API requests and responses using the Network tab in Browser DevTools.
- I have developed skills to debug and optimize React applications for better performance and reliability.